

Research Proposal

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Title

Thermodynamic Stabilization of Ni Catalysts via Promoter-Defect Synergy for Propane Dehydrogenation under Regeneration Conditions

Abstract

Propylene serves as a fundamental building block for polymers and high-value chemicals, driven by a consistently escalating global demand. While propane dehydrogenation (PDH) offers a strategic on-purpose route for propylene production, industrial applications remain confined to noble-metal Pt or toxic Cr-based systems due to the structural vulnerability of earth-abundant alternatives. Ni-based catalysts emerge as a compelling alternative owing to their cost-effectiveness and superior C-H bond activation efficiency. However, the excessive hydrogenolysis activity of metallic Ni ensembles often triggers C-C bond cleavage, leading to undesired methane formation and rapid coking, which severely undermines selectivity and catalyst life.

This research proposes a regeneration-aware design strategy to stabilize isolated Ni active sites through the synergistic integration of promoter-induced electronic modulation and defect-anchored metal-support interactions. Utilizing boron (B) as a model promoter alongside defect-engineered oxide supports, this work aims to construct a thermodynamically robust coordination environment capable of suppressing Ni sintering and ensemble re-formation during harsh, repetitive PDH-regeneration cycles. By elucidating the dynamic structure-activity-stability relationships, this study seeks to transcend the intrinsic limitations of Ni-based catalysis and establish a universal framework for designing stable, non-precious metal catalysts tailored for industrially relevant redox environments.

Keywords: Propane dehydrogenation; Ni-based catalysts; Promoter-support synergy; Defect engineering.

1. Research Background and Literature Review

Propylene is the basic raw material for the production of acrolein, polypropylene, acetone, acrylonitrile, propylene oxide, and other industrial products.¹ With the increasing availability of propane, propane dehydrogenation (PDH, $\text{C}_3\text{H}_8 \rightleftharpoons \text{C}_3\text{H}_6 + \text{H}_2$) has become an important method for producing propylene, particularly as traditional steam cracking

becomes increasingly constrained by feedstock flexibility and carbon intensity.^{2,3} Currently, industrial PDH applications are dominated by two flagship technologies: the UOP Oleflex process, which utilizes platinum-based catalysts (e.g., Pt-Sn/Al₂O₃), and the ABB Lummus Catofin process, which employs chromium-based catalysts (e.g., Cr₂O₃/Al₂O₃). However, these established systems face persistent challenges: the Oleflex process relies on costly noble metals and is sensitive to feed impurities, while the Catofin process is hampered by the environmental toxicity of chromium and the inherent deactivation caused by pronounced coke formation during high-temperature cycles.^{1,4} To address these limitations, significant research efforts have been directed toward developing cost-effective and environmentally benign oxide catalysts. However, the development of such substitutes has been impeded by an insufficient understanding of how the active site structure evolves during repeated high-temperature reaction regeneration cycles, which is a key factor governing catalyst stability and selectivity during cyclic PDH operation.

Ni-based catalysts are appealing candidates owing to their low cost and strong capability for C-H bond activation.⁵ However, the perceived unsuitability of Ni for PDH primarily stems from its inability to maintain isolated active sites under high-temperature reaction and regeneration cycles, rather than from insufficient activity in C-H activation itself. Consequently, the primary catalytic challenge manifests not as overcoming intrinsic activation barriers, but rather as circumventing selectivity attenuation and structural degradation under industrially relevant thermal and redox regimes. Under dehydrogenation conditions, contiguous metallic Ni ensembles readily catalyze C-C bond scission reactions, leading to methane formation, excessive hydrogenolysis, and rapid coke accumulation.^{6,7} These processes result in poor propylene selectivity and fast deactivation, especially during the high-temperature oxidative-reductive regeneration cycles required for industrial PDH operation.

Although various strategies have been explored to address these issues, including doping (e.g., NiP⁸, NiSn^{9,10}, NiAu¹¹), vapor-phase treatment¹², defect engineering², and organometallic synthesis, superior catalytic performance remains elusive. This is primarily because these approaches still rely on Ni⁰ as the active phase, which suffers from inherent limitations in selectivity and stability. Consequently, preventing the dynamic reduction and aggregation of Ni species into contiguous metallic ensembles under reaction-regeneration cycling has emerged as the central unresolved challenge.

Nevertheless, most of the reported Ni separation strategies have been evaluated under static or weak reduction conditions, while the thermodynamic properties and change of Ni sites during the repeated oxidation-reduction regeneration process have not been fully studied.

2. Central Hypothesis and Research Objectives

This work hypothesizes that the long-standing failure of Ni catalysts in PDH does not arise from insufficient intrinsic C-H activation capability, but rather from the lack of a thermodynamically stable coordination environment capable of suppressing Ni ensemble formation and re-formation under harsh oxidative-reductive cycling conditions.

Specifically, this work hypothesizes that stabilizing Ni active sites under PDH conditions requires the simultaneous satisfaction of both electronic and geometric constraints; neither constraint alone is sufficient to ensure stability. Electron-withdrawing promoters such as boron are hypothesized to increase the reduction barrier of Ni species by stabilizing partially oxidized Ni centers, while defect-engineered oxide supports can provide strong anchoring sites that suppress surface diffusion and aggregation. The synergistic coupling of electronic modulation and defect anchoring is proposed to enhance thermodynamic stability against ensemble formation. This hypothesis will be evaluated against control systems with a single modification.

Based on this hypothesis, the following testable predictions are proposed:

1. Stabilized isolated Ni sites are expected to retain a low apparent Ni-Ni coordination number after oxidative regeneration, remaining significantly below that of conventional Ni catalysts and approaching zero within experimental uncertainty.
2. Electronic modification of Ni sites weakens the adsorption of propylene, thereby promoting its rapid desorption and suppressing secondary reactions leading to coking.
3. The suppression of metal component aggregation during regeneration is hypothesized to directly enhance propylene selectivity and maintain catalytic stability under cyclic PDH conditions.

The objective of this study is to establish a structure-activity relationship for regenerative catalysts, linking the Ni coordination environment, electronic structure, and catalytic performance, thus providing design principles for rational development of non-precious metal dehydrogenation catalysts.

3. Research Questions

This proposal is structured around the following hypothesis-driven research questions:

1. How does promoter-induced electronic modulation influence the oxidation state and ensemble size of Ni species under PDH conditions?

This question will be addressed by operando X-ray absorption spectroscopy (XAFS) analysis of Ni valence state and apparent Ni-Ni coordination number, complemented by probe-molecule IR spectroscopy to assess electronic perturbation.

2. What role do support surface defects and acid-base properties play in anchoring Ni species and preventing sintering during high-temperature regeneration?

Correlations between defect density, acid-base characterization, and Ni dispersion after cycling will provide direct evidence for anchoring effects.

3. How do stabilized Ni active sites modify propylene adsorption geometry and reaction pathways, thereby suppressing deep cracking and coke formation?

Propylene adsorption configurations will be examined by in situ DRIFTS, while coke precursors and carbon species evolution will be monitored using Raman spectroscopy and temperature-programmed oxidation.

4. How does the dynamic structural evolution of Ni sites under reaction conditions correlate with catalytic performance metrics such as conversion, selectivity, and stability?

Time-resolved operando spectroscopy will be combined with catalytic cycling tests in a time-resolved operando setup to establish structure-performance relationships.

4. Research Originality and Significance

This work differs from conventional Ni catalyst modification strategies that rely on alloying or empirical promoter screening. Rather than transiently diluting Ni ensembles, the proposed approach aims to achieve thermodynamic stabilization of isolated Ni active sites under industrially relevant PDH conditions.

The innovation of this research manifests in three key aspects. First, it explicitly considers the catalyst support as a non-innocuous component, whose defect structure and acid-base properties decisively influence catalyst stability and selectivity. The validity of this framework relies on establishing quantitative correlations between support defect density, intrinsic acid-base properties, and post-regeneration Ni dispersion and propylene selectivity trends. Second, the study focuses on suppressing the reformation of Ni clusters during regeneration through defect engineering and electronic regulation—a factor often overlooked in research targeting only fresh or lightly treated catalysts. This hypothesis will be validated if in situ XAFS techniques reveal that Ni coordination numbers remain persistently low and catalytic selectivity remains stable after repeated redox cycles. Third, it employs in situ and operando spectroscopy to directly monitor active site evolution and reaction intermediates, thereby constructing a dynamic structure-activity relationship instead of a static one. This framework will gain empirical support if time-resolved structural descriptors can quantitatively correlate with transient changes in conversion, selectivity, and coke formation.

The insights gained from this study will not only deepen fundamental understanding of Ni-catalyzed alkane dehydrogenation but also provide generalizable design principles to guide

the development of robust, low-cost catalysts for high-temperature hydrocarbon transformations.

5. Research Methodology

5.1 Catalyst Design and Synthesis

Taking boron as an example, it was selected as a model promoter due to its strong electron-withdrawing properties and its ability to modulate surface acidity while avoiding the formation of bulk metallic alloys with Ni. To decouple geometric isolation from electronic promotion, a strong electrostatic adsorption (SEA) scheme will be employed to precisely anchor the Ni precursor at defect sites near boron-modified regions. By adjusting the precursor solution pH relative to the carrier's zero-charge point, we will selectively target defect sites on the carrier surface to ensure electrostatic anchoring of the Ni precursor.

The pH and ionic strength of impregnation solutions will be carefully controlled to promote the electrostatic adsorption of precursors onto defect sites of the support while avoiding premature precipitation of Ni species.

Given the instability of boron species at elevated temperatures, a stepwise calcination protocol will be developed, incorporating a controlled heating profile and intermediate soak steps. This approach ensures complete decomposition of the precursor while minimizing loss of catalyst additives and preserving the intimate interaction between Ni and B. Quantitative assessment of additive retention after reaction-regeneration cycles will be performed via bulk elemental analysis (ICP-OES) and XPS measurements, thus distinguishing overall boron loss from surface redistribution effects.

5.2 Catalytic Performance Evaluation

The PDH performance of catalysts will be evaluated in a fixed-bed quartz reactor under both differential and integral conditions. Online gas chromatography will be employed to quantitatively measure propane conversion, propylene selectivity, and byproduct distribution. Long-term stability tests and repeated reaction-regeneration cycles will be performed to evaluate anti-sintering and anti-coking properties.

5.3 Structural and Chemical Characterization

Standard bulk crystallographic analysis via XRD will be augmented by XAFS. Specifically, EXAFS modeling will be used to quantitatively determine coordination numbers and interatomic distances in the Ni-O and Ni-B coordination shells. Furthermore, probe-molecule infrared spectroscopy, utilizing CO and NO as adsorbates, will be used to rigorously analyze the localized electronic states of surface-active Ni species.

To elucidate reaction mechanisms, in situ diffuse reflectance infrared Fourier transform spectroscopy and in situ Raman spectroscopy will be used to track adsorbed hydrocarbon species and coke precursors under reaction conditions. These measurements will directly link surface chemistry to catalytic performance and test the proposed hypotheses regarding adsorption geometry and coke suppression.

Each characterization technique employed in this work is explicitly designed to test a specific aspect of the central hypothesis rather than to provide descriptive structural information alone. (XAFS will directly quantify the coordination environment and ensemble size of Ni species, thereby evaluating the effectiveness of defect anchoring and promoter stabilization under both reaction and regeneration conditions. In particular, changes in Ni-Ni coordination numbers before and after oxidative regeneration will serve as a decisive metric for ensemble suppression.

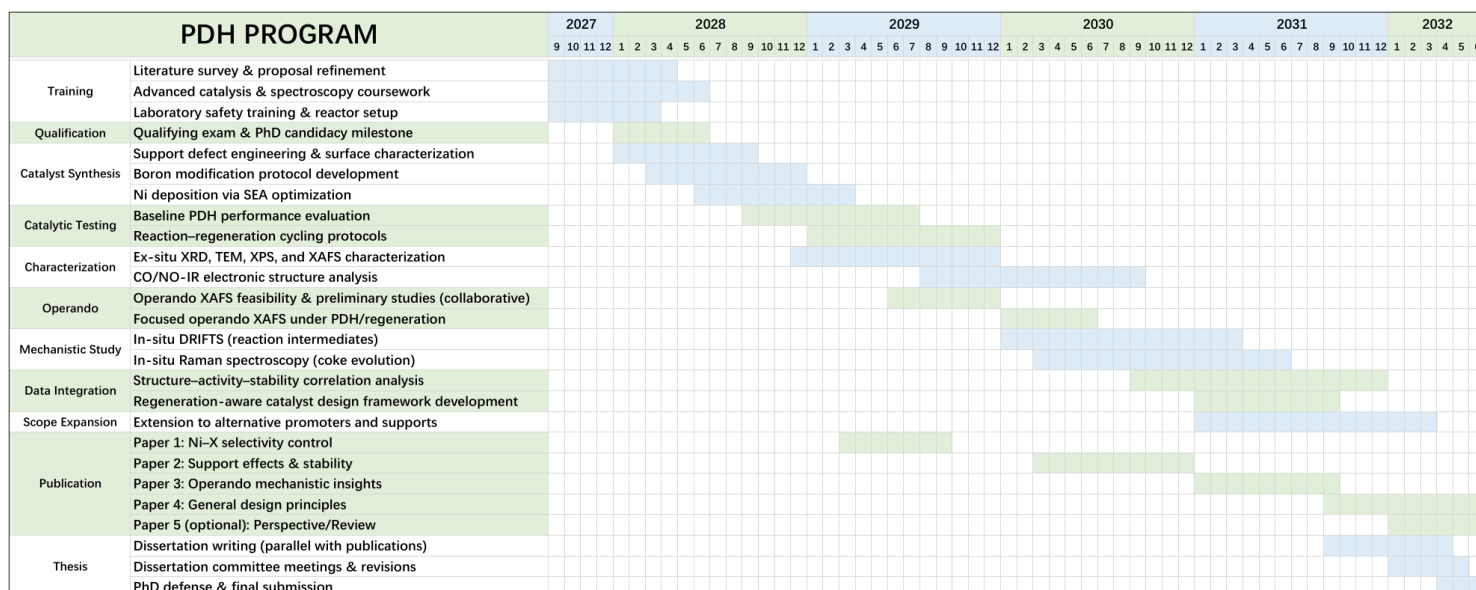
Probe-molecule infrared spectroscopy using CO and NO will be employed to interrogate the electronic structure and surface accessibility of Ni sites. Shifts in adsorption frequency and binding configuration will be correlated with promoter-induced electronic effects and their impact on propylene adsorption.

In situ DRIFTS and Raman spectroscopy will provide mechanistic insights into reaction intermediates and coke precursor evolution under working conditions. By correlating surface species evolution with catalytic performance and structural stability, this work aims to construct a dynamic structure-activity-stability relationship that goes beyond static catalyst descriptors.

6. Expected Outcomes

This research is expected to achieve the following measurable outcomes: 1. To prove that under PDH conditions, by combining the assistance of additives with the anchoring of carrier defects, isolated Ni sites can be stabilized. Specifically, after repeated redox cycles, the apparent Ni-Ni coordination number remains low, and propylene selectivity decay remains negligible during 50–100 h stability testing. 2. By establishing a quantitative correlation between the size of Ni clusters, the adsorption characteristics of propylene, the reduction rate of carbon precursors, and the generation rate of graphite carbon substances, to clarify the mechanism underlying the enhancement of selectivity and the inhibition of carbon formation. 3. To define the transferable design standards applicable to non-precious metal catalysts in extreme thermal and redox environments, expressed in recognizable structural descriptors, to predict stable catalytic performance over multiple regeneration cycles. 4. To establish a relevant model for the addition of Ni-based catalyst additives.

7. Research Timeline



8. Summary and Outlook

By addressing the thermodynamic instability of Ni active sites during cyclic reaction-regeneration conditions, this study aims to fundamentally redefine the role of Ni in the propane dehydrogenation reaction. It does not view Ni as a metal inherently unsuitable, but rather redefines its limitations as solvable problems related to coordinated environment control and dynamic structural stability, thereby improving the catalyst model with Ni as the active metal.

The regeneration-aware design principles established in this study are expected to be broadly applicable to other earth-abundant metals and high-temperature hydrocarbon transformations, providing a general framework for bridging the gap between academic catalyst design and practical industrial operation.

9. Reference

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